

Intro & Concepts

Data Visualization

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Data Types

1. Categorical Variables (Qualitative Data)

a) Nominal Variables

- **No natural order or ranking**
- Examples: *colors, gender, country, blood type*
- Can be represented as *dummy variables* (0/1 encoding)
- Can be *Boolean variables* (True/False, Yes/No)

```
african_countries
```

```
"$150K+"
)
```

2. Numerical Variables (Quantitative Data)

a) Discrete Variables

- Countable, finite values
- Examples: *number of children, exam scores, count of items*
- Often integers but not always

Student Name	Score
Abigail Gonzalez	85
Alexia Jones	69
Alyssa Lopez	67
Amanda Hernandez	82
Amelia Hernandez	88
Andrew Gonzalez	65
Andrew Lopez	82
Anthony Garcia	89
Ariana Sanchez	78
Ashley Johnson	75

3. Time-Based Data

a) Time Series Data

- Data points indexed by time
- Examples: *stock prices, weather data, sales over time*

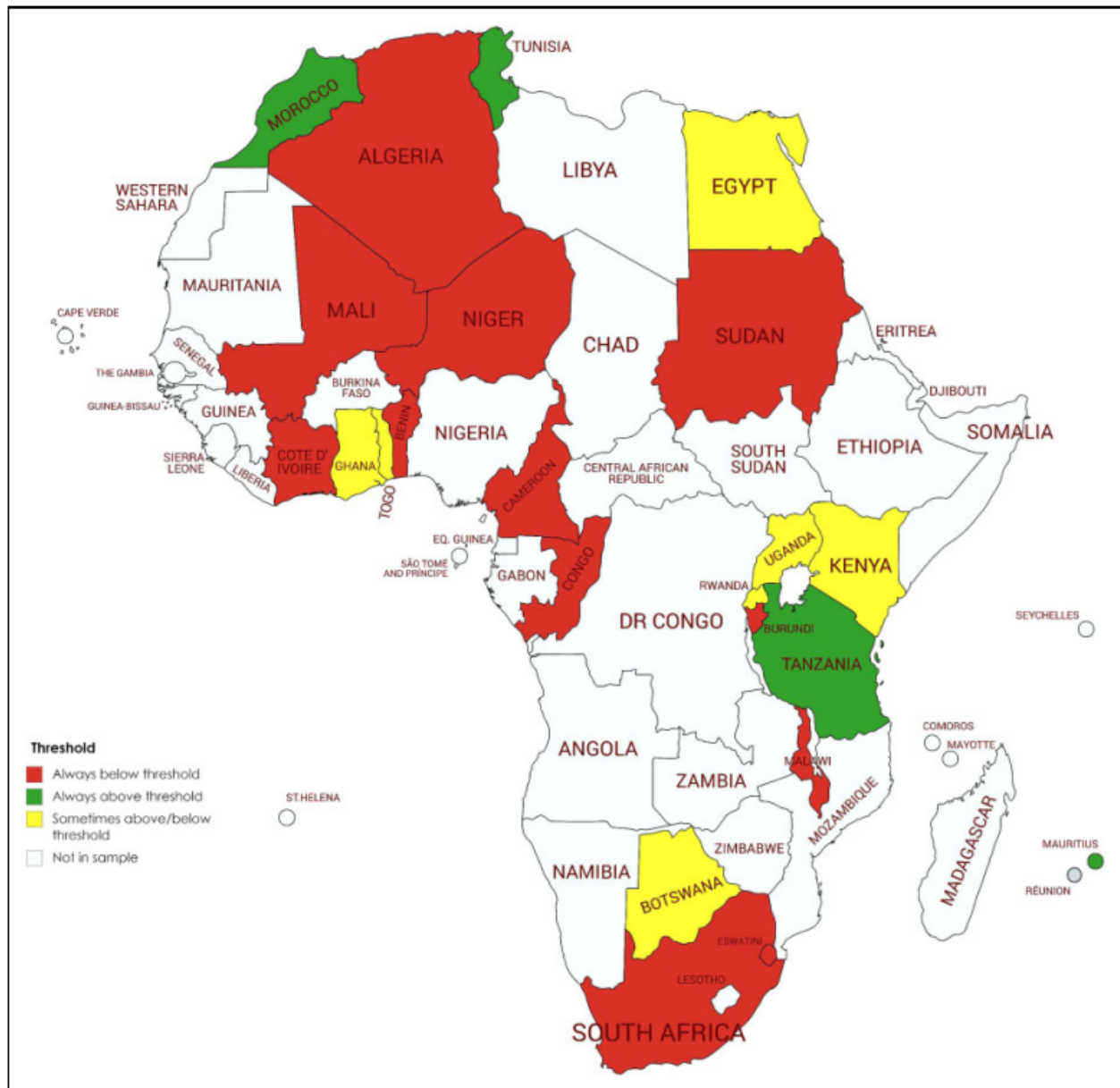
date	open_price	high_price	low_price	close_price	volume
1/3/2011	26.84	27	25.9	26.62	1283000
1/4/2011	26.66	26.95	26.02	26.67	1187400
1/5/2011	26.48	26.9	26.19	26.83	1446700
1/6/2011	2				

Country	Year	GDP per Capita	Unemployment	Inflation	Govt Debt/GDP	Trade Balance/GDP
Australia	2021	\$59,286	3.5%	5.1%	65.0%	-1.7%
Australia	2022	\$59,269	3.2%	4.7%	67.1%	-1.8%
Australia	2023	\$62,020	4.0%	2.7%	73.8%	-0.9%
Australia	2024	\$61,614	4.7%	2.4%	72.6%	-0.2%
Canada	2015	\$46,320	6.3%	1.8%	86.7%	0.7%
Canada	2016	\$47,124	7.7%	3.8%	89.9%	2.9



b) Polygon Data

- Areas or regions
- Examples: *country boundaries, sales territories*



c) Raster Data

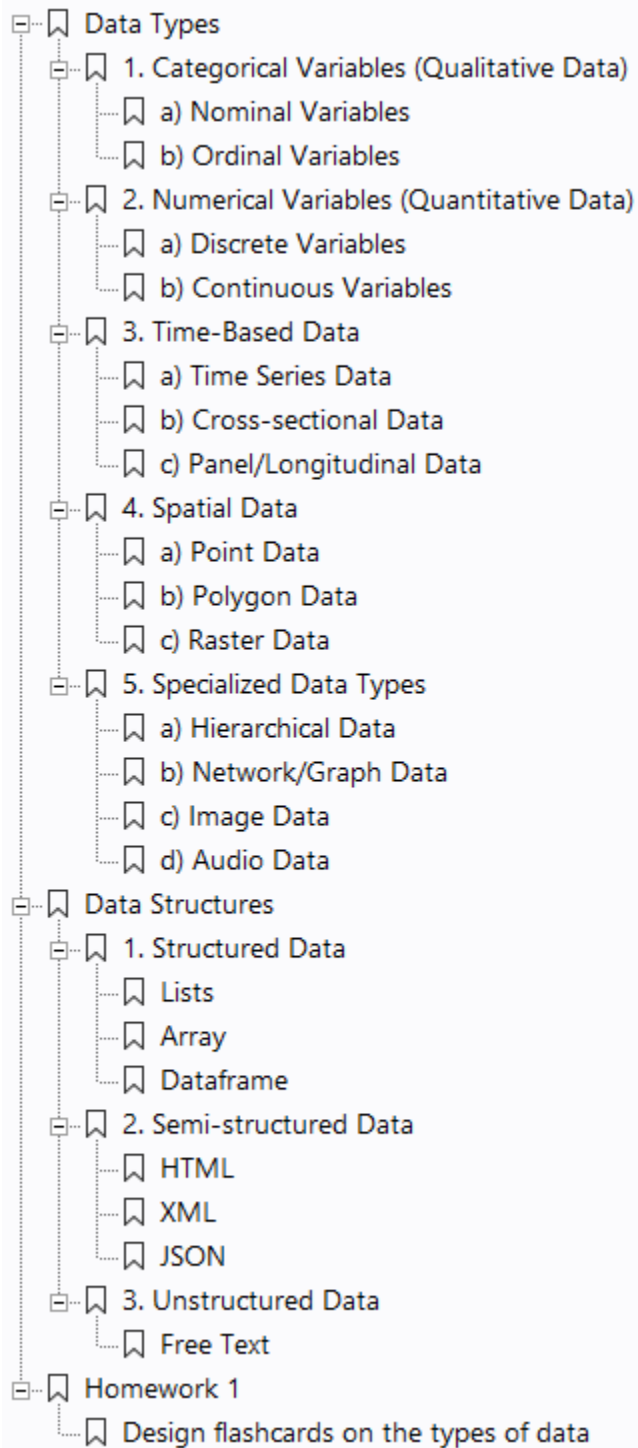
- Grid-based spatial data
- Examples: *satellite imagery, elevation maps*



5. Specialized Data Types

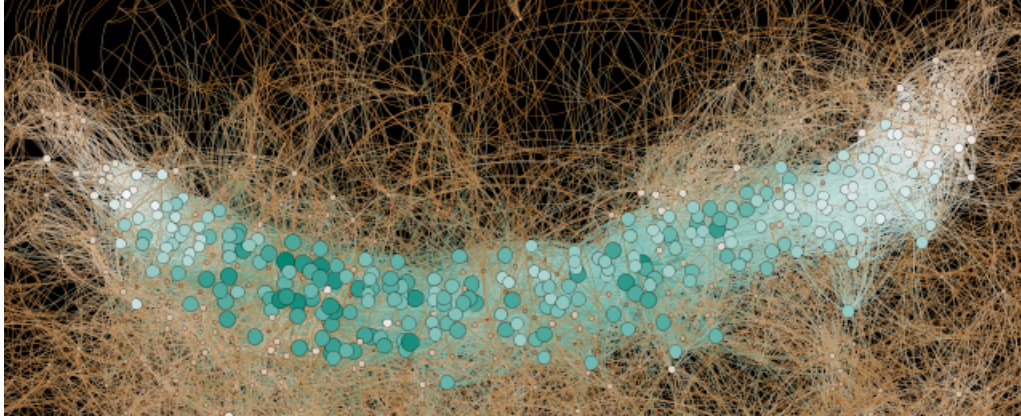
a) Hierarchical Data

- Tree-like structures
- Examples: *organizational charts, file systems*



b) Network/Graph Data

- Nodes and edges representing relationships
- Examples: *social networks, transportation systems*



c) Image Data

- Pixel-based visual information
- Can be *grayscale*, *RGB*, or *multispectral*



d) Audio Data

- Sound waves represented digitally
- Time-series of amplitude values



Data Structures

1. Structured Data

Lists

Collection of similar items

```
my_list <- list(  
  graduate_class_level = TRUE,  
  number_of_students = 24,  
  first_names = c(  
    "Alexander", "Sophia", "Benjamin", "Isabella", "Christopher", "Emma",  
    "Daniel", "Olivia", "Ethan", "Ava", "Gabriel", "Charlotte", "Henry",  
    "Amelia", "Jackson", "Harper", "Liam", "Evelyn", "Matthew", "Abigail",  
    "Nathan", "Emily", "Samuel", "Victoria"  
  )  
)
```

Array

Two-dimensional arrangement of numerical values optimized for mathematical operations

```
my_array <- array(1:12, dim = c(3, 4))
#      [,1] [,2] [,3] [,4]
# [1,]  1   4   7  10
# [2,]  2   5   8  11
# [3,]  3   6   9  12
```

Dataframe

2D table structure (like Excel spreadsheet)

	country	gdp_2015	gdp_2024	growth_rate
1	South Korea	31643	45992	45.3
2	UK	41499	51526	24.2
3	USA	54607	65689	20.3
4	Australia	52015	61614	18.5
5	Germany	47655	56386	18.3
6	France	41343	48551	17.4
7	Canada	46320	54189	17.0
8	Japan	41099	47173	14.8

2. Semi-structured Data

HTML

Text with some formatting or tags: *HTML tags*

```
<table border="1">
  <thead>
    <tr>
      <th>ID</th>
      <th>Name</th>
      <th>Department</th>
      <th>Salary</th>
      <th>Status</th>
    </tr>
  </thead>
  <tbody>
    <tr>
      <td>001</td>
      <td>John Smith</td>
      <td>Engineering</td>
      <td>$75,000</td>
```

```

        <td>Active</td>
    </tr>
    <tr>
        <td>002</td>
        <td>Sarah Johnson</td>
        <td>Marketing</td>
        <td>$65,000</td>
        <td>Active</td>
    </tr>
    <tr>
        <td>003</td>
        <td>Mike Davis</td>
        <td>Sales</td>
        <td>$55,000</td>
        <td>Inactive</td>
    </tr>
    <tr>
        <td>004</td>
        <td>Emily Brown</td>
        <td>HR</td>
        <td>$60,000</td>
        <td>Active</td>
    </tr>
    <tr>
        <td>005</td>
        <td>David Wilson</td>
        <td>Engineering</td>
        <td>$80,000</td>
        <td>Active</td>
    </tr>
</tbody>
</table>

```

ID	Name	Department	Salary	Status
001	John Smith	Engineering	\$75,000	Active
002	Sarah Johnson	Marketing	\$65,000	Active
003	Mike Davis	Sales	\$55,000	Inactive
004	Emily Brown	HR	\$60,000	Active
005	David Wilson	Engineering	\$80,000	Active

XML

Text with some formatting or tags: *XML tags*

```

<bookstore>
  <book category="children">
    <title>Harry Potter</title>
    <author>J K. Rowling</author>
  </book>
</bookstore>

```



```

    <year>2005</year>
    <price>29.99</price>
</book>
<book category="web">
  <title>Learning XML</title>
  <author>Erik T. Ray</author>
  <year>2003</year>
  <price>39.95</price>
</book>
</bookstore>

```

JSON

Text with some formatting or tags: *JSON*

```

{
  "SPX/USD:1750469492357": {
    "return": {
      "period_1hr_return": 0.008154943934760572,
      "period_4hr_return": -0.017076255872786383,
      "period_6hr_return": -0.029873372570001666,
      "period_12hr_return": -0.15160258909771496,
      "period_24hr_return": -0.16076525495641436,
      "period_36hr_return": -0.20539040245416684,
      "period_48hr_return": -0.2463978941535051,
      "period_72hr_return": -0.17826119797567785
    },
    "volume": 8479316.84196,
    "ret_ts": 1750469492357,
    "ret_time": "20 June, 2025 Friday 21:31:32 PM"
  },
  "UNI/USD:1750469793451": {
    "return": {
      "period_1hr_return": 0.003966005665722472,
      "period_4hr_return": -0.00603000981629509,
      "period_6hr_return": -0.003514691410094195,
      "period_12hr_return": -0.08234075608493008,
      "period_24hr_return": -0.09707006369426752,
      "period_36hr_return": -0.06466086038532592,
      "period_48hr_return": -0.06218576342947868,
      "period_72hr_return": -0.04060638873849487
    },
    "volume": 482739.58039564,
    "ret_ts": 1750469793451,
    "ret_time": "20 June, 2025 Friday 21:36:33 PM"
  }
}

```

3. Unstructured Data

Free Text

- Natural language content

- Examples: *reviews, emails, documents*

Customer called on March 15, 2024 at 2:30 PM regarding order #12847. They were unhappy with the delayed shipping (originally promised 3-5 business days, but it's been 8 days). Customer's email is sarah.johnson@email.com and phone number is 555-0123. They spent \$127.50 on 3 items: wireless headphones (\$89.99), phone case (\$19.99), and charging cable (\$17.52). Customer satisfaction rating would be 2/10. They mentioned they're a premium member since 2019 and expect better service. Promised to escalate to supervisor and provide 15% discount on next order. Follow-up required by March 18th.

Data you can extract

Field	Value
Date	March 15, 2024
Time	2:30 PM
Order number	12847
Email	sarah.johnson@email.com
Phone	555-0123
Total amount	\$127.50
Rating	2/10
Member since	2019
Discount	15%
Customer sentiment	unhappy

Homework 1

Design flashcards on the types of data

- several data types were introduced today
 - *categorical, numerical ... etc.*
- create flashcards
 - individual assignment—you will be building on this for the rest of the term
 - PDF files, one data type per file
 - with the name of the data type on the first page (e.g., *categorical* data)
 - with enough information for you to remember what it is—description, salient features, examples/illustrations on the second page
 - be creative, but stick to the task
- grade based on *accuracy* of information and *efficiency* of real estate used
- upload in canvas